figure1A | native width; print at 100%

Case	Local indicator	Realized outcome
M4 HumanEval30, k = 2	V_2^{micro}: 1.022–1.076	Profitable: 1.387×; 90/90 exact
M4 representative 24Q	V_2^{micro}: 1.022–1.076	Negative: 0.842×; 72/72 exact
RTX 5070 Ti Laptop, batched	V_2^{micro}: 0.983–1.009	Blocked; no admissible speedup
358H NPU draft	NPU/CPU: 48.08 vs 30.81 tokens/s	Negative: 0.352×; 90/90 exact

Favorable local signals, divergent realized outcomes. Near-one-step target batching and favorable isolated drafting do not consistently translate into qualified end-to-end gains in the tested cases. The same tested M4 Metal stack exhibits a profitable HumanEval regime and a negative representative-24Q regime in separate frozen studies. RTX batched verification is physically cheap but fails correctness qualification. The 358H NPU has higher isolated median throughput at 8,192 tokens yet reaches only 0.352× realized speedup. Rows are heterogeneous case summaries, not a hardware ranking or common quantitative axis. M4 micro values describe the shared path in a separate micro study, not independent measurements in each workload. The 8,192-token NPU example is contextualized by all three contexts in the 358H figure. Headline speedups cover post-prefill decode; the M4, verification, and 358H figures give conditions, uncertainty, and the two measured diagnoses.

figure1B_table | native width; print at 100%

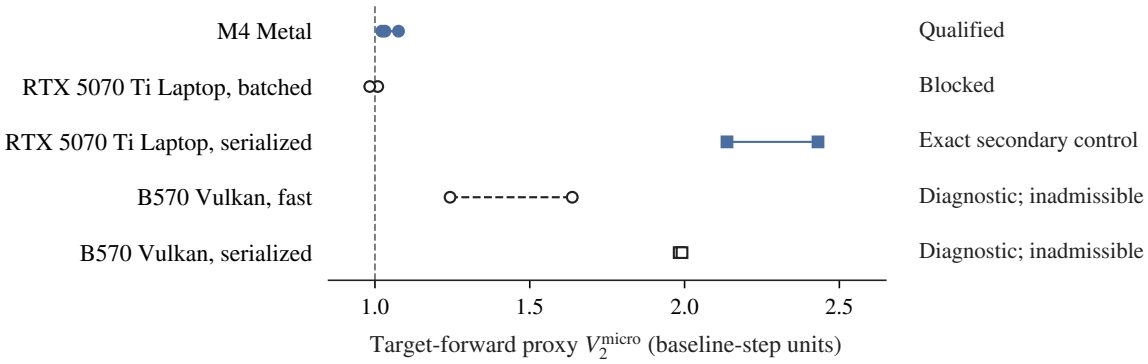
Case	Local indicator	Realized outcome
M4 HumanEval30, $k = 2$	$V_2^{\text{micro}} = 1.022$–$1.076$	Profitable: 1.387×; 90/90 exact
M4 representative 24Q	$V_2^{\text{micro}} = 1.022$–$1.076$	Negative: 0.842×; 72/72 exact
RTX 5070 Ti Laptop, batched	$V_2^{\text{micro}} = 0.983$–$1.009$	Blocked; no admissible speedup
358H NPU draft	NPU/CPU: 48.08 vs 30.81 tokens/s	Negative: 0.352×; 90/90 exact

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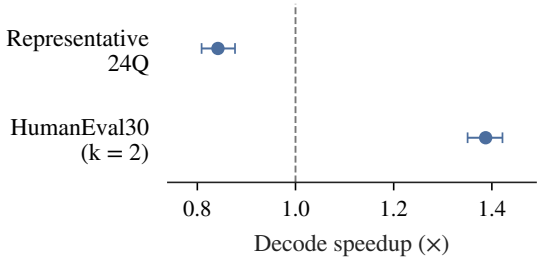
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The shared M4 Metal path has a separately measured V_2^{micro} range of 1.022–1.076, yet the separate HumanEval30 ($k = 2$) and representative-24Q studies yield 1.387× (90/90 exact) and 0.842× (72/72 exact). RTX batched verification costs 0.983–1.009 baseline steps but fails qualification, so it has no admissible speedup. At 8,192 tokens, isolated NPU/CPU drafting reaches 48.08/30.81 tokens/s, while the qualified 358H NPU path yields 0.352× (90/90 exact). Speedups cover post-prefill decode; the detailed figures retain campaign-specific intervals and measurement boundaries. These heterogeneous observations do not isolate workload-only causality.

verification | native width; print at 100%

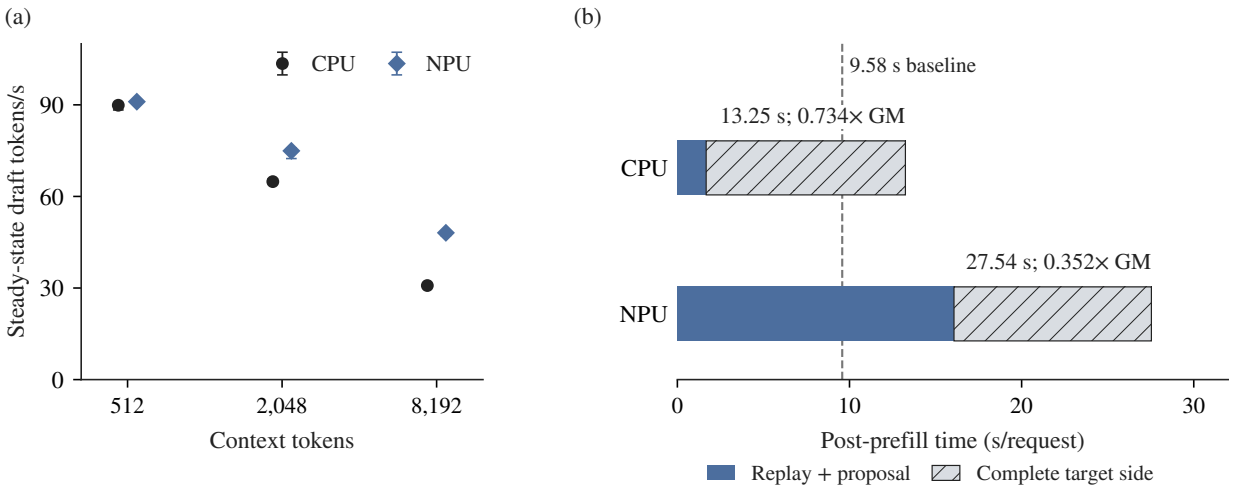


Correctness qualification changes which verification path is usable. Each mark is one measured context cell; range segments span those cells, not confidence intervals. Filled markers indicate qualified tested paths; open markers indicate blocked or diagnostic paths. Serialization yields an exact secondary RTX control, but the B570 serialized diagnostic remains globally inadmissible despite improving observed exactness from 45/90 to 60/90 pairs. Contexts and runtime stacks are not fully matched. The target-forward proxy V_2^{micro} is not complete realized target-side cost R_{target} or an end-to-end speedup. Qualification is scoped to the frozen workload and execution path, not a device family. M4 contexts are 512/2,048/8,192 tokens, RTX depths 0/256, and B570 contexts 512/2,048; nearby cells may overlap. Circles denote batched/fast paths and squares serialized paths. Ranges are not CIs; $V_2^{\text{micro}} \neq R_{\text{target}}$.



Opposite profitability regimes on the same tested M4 stack. Representative 24Q yields 0.842× [0.809, 0.877], while the separately preregistered HumanEval30 fixed-$k = 2$ study yields 1.387× [1.351, 1.422]. Points are campaign prompt geometric means of paired decode speedups; intervals are campaign-specific 95% prompt-bootstrap intervals. The dashed line marks parity. Both paths satisfy token equality (72/72 and 90/90 pairs), which is not functional HumanEval pass@k. These are separate studies, not a paired between-workload comparison; differing workloads and session conditions do not isolate workload-only causality.

state | native width; print at 100%



Isolated draft advantage does not survive the measured stateful integration. (A) Phase-1 median steady-state draft throughput and 95% repetition-bootstrap intervals, five repetitions per cell; the NPU median advantage grows across the tested contexts. (B) Phase-3 arithmetic mean post-prefill seconds per request over 90 observations per path, all exact. CPU and NPU totals are 13.25 s and 27.54 s against a 9.58 s mean baseline. The CPU path is target-side dominated (87.34%); the NPU path spends 58.35% in the measured replay-plus-proposal call. The 0.734× and 0.352× headlines use the separate prompt-GM estimator and are not ratios of the arithmetic mean bars. Replay and proposal share one timed OpenVINO call; replay-only time is not measured. Initial prompt prefill is excluded. This diagnoses the tested runtime interface, not intrinsic NPU capability or a measured benefit from a state-persistent intervention. Together with the verification figure, it identifies two measured reasons local opportunity need not become realized gain; the M4 contrast remains a separate empirical observation. CPU replay plus proposal is 12.63%; NPU complete target work is 41.63%. Residuals are 4.81 ms (0.04%) and 5.06 ms (0.02%). Prompt-GM speedup 95% CIs are [0.693, 0.771] and [0.333, 0.369], respectively. The vertical dashed line is the shared arithmetic-mean baseline. Isolated throughput excludes the full-prefix state cost; its device marks are horizontally offset for readability.

appendix_table | native width; print at 100%

Case	Qualification	Outcome	Class	Case annotation
M4 representative 24Q	Qualified; 72/72 exact	Negative	P	Workload-conditioned
M4 HumanEval30	Qualified; 90/90 exact	Profitable	P	Workload-conditioned
RTX batched	Blocked; gate failed	Not admissible	M	Verification-constrained
RTX serialized controls	Qualified; 90/90 each exact	Negative	S	Verification-constrained
B570 fast	Inadmissible; 45/90 exact	Not admissible	D	Correctness-blocked
B570 serialized	Inadmissible; 60/90 exact	Not admissible	D	Correctness-blocked
358H CPU	Qualified; 90/90 exact	Negative	P/M	Target-side dominated 87.34% target side
358H NPU	Qualified; 90/90 exact	Negative	P/M	State-interface bound 58.35% replay + proposal

Evidence inventory across tested regimes. Categorical columns distinguish globally blocked paths from exact negative and profitable regimes. Rows annotate observed constraints rather than define an exhaustive taxonomy. Both B570 paths remain diagnostic and globally inadmissible. RTX serialized controls are secondary evidence; 358H outcome and decomposition support primary and mechanism claims, respectively. Empty cells indicate no displayed case, not impossibility. No hardware ordering or continuous correctness metric is implied. P: primary; S: secondary; M: mechanism; D: diagnostic. RTX serialized groups CPU-to-RTX and RTX-to-RTX controls, 90/90 exact pairs each. Fractions denote generated-token equality, not functional correctness.